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Integrated Ferroelectrics
An International Journal

ISSN: 1058-4587 (Print) 1607-8489 (Online) Journal homepage: http://www.tandfonline.com/loi/ginf20

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To cite this article: SeongMin Kim, Jaewook Ha & Jin-Baek Kim (2017) Morphology effect on the transferred charges in triboelectric nanogenerators: Numerical study using a finite element method, Integrated Ferroelectrics, 183:1, 19-25, DOI: 10.1080/10584587.2017.1375820

To link to this article: https://doi.org/10.1080/10584587.2017.1375820

Published online: 11 Dec 2017.

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INTEGRATED FERROELECTRICS
2017, VOL. 183, 19–25
https://doi.org/10.1080/10584587.2017.1375820

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Morphology effect on the transferred charges in triboelectric nanogenerators: Numerical study using a finite element method

SeongMin Kim[a,+], Jaewook Ha[b,+], and Jin-Baek Kim[b]

[a]The Office of Computational Energy Science (private lab), Gyeonggi-do, Korea; [b]Department of Chemistry, Korea Advanced Institute of Science and Technology, Daejeon, Korea

ABSTRACT

The structural shape of the interface between a metal and dielectric material in a triboelectric nanogenerator (TENG) is an important factor that can improve the device performance. Many interfacial structures have been developed to improve the TENG performance. However, there have been very few studies on the numerical interpretation of various types of contact interfaces. For various interfacial structures on which uniform triboelectric charge density is distributed, the surface charge density (in-plane, out-of-plane, and total) is systematically analyzed to predict the quantity of the transferred charges on the bottom metal under a short-circuit condition. In this work, a numerical study is conducted using a finite element method. The numerical results confirm that the increase in the quantity of the transferred charges collected on the bottom metal via electrostatic induction is related to the increase in the area of the surface structures (i.e., surface enlargement effect due to the formation of complex interfacial morphology). The estimated magnitude of the transferred charges shows the following decreasing trend for the various structural shapes: rectangle > cylinder > pyramid > cone > flat.

ARTICLE HISTORY
Received 12 October 2016
Accepted 24 April 2017

KEYWORDS
Interfacial structures; transferred charges; triboelectric nanogenerators; FEM

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1. Introduction

The triboelectric charge density on the surface of a dielectric material can be increased via a geometrical approach, which involves increasing the contact area via nano- or microstructuring as the contact area can be controlled by an increase in the area of the surface structures. This is referred to as interfacial surface engineering [1–14], and it has been used to increase the triboelectric nanogenerator (TENG) device performance. Another way to increase the triboelectric charge density is to use appropriate materials [15]. Triboelectric materials with a large difference between their work functions are selected [15–16] because this can make the

CONTACT S. M. Kim smkim417@gmail.com Samsung Advanced Institute of Technology, Suwon-si, Gyeonggi-do 443-803, Korea.
+Equal contribution.
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electron accepting or donating material sufficiently strong to maximize the electron transfer process between the two triboelectric materials [16]. Patterned interfaces of various shapes have already been considered and studied before. For example: Line, pillar and hexagonal cone-shapes are discussed by M. A. Parvez Mahmud et al. [17], nanopillar arrays are studied by Weixing Song et al. [18], cylindrical/spiral shaped interfaces are described by L. Xiao-Hiu et al. [19], and information on general textured interfaces can be found in Cheng Guang-Gui et al. [20]. In this work, the surface charge density distribution on various structural shapes, such as a cone, pyramid, rectangle, cylinder, and flat, is analyzed systematically and comprehensively using a finite element method (FEM). The numerical results indicate that there is a correlation between the amount of transferred charges in a TENG and interfacial morphology. Therefore, interfacial design is important for TENG applications.

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2. Numerical methods: Finite element method with interfacial structures

Five interfacial morphologies (cylindrical, conical, rectangular, pyramidal, and flat structures) were analyzed using an FEM to calculate the induced surface charge density from the electric displacement field distribution, as shown in Fig. 1. The dimensions of the model structures used in the simulation are as follows: The area of the basal plane is 60 × 60 μm. The pitch and height of each structure (cylinder, cone, rectangle, and pyramid) is 20 μm and 10 μm, respectively. The thickness of the top and bottom electrode (Al) is 2 μm. The gap between the top electrode and the dielectric structure is filled with air and has a thickness of 23 μm. Each structure

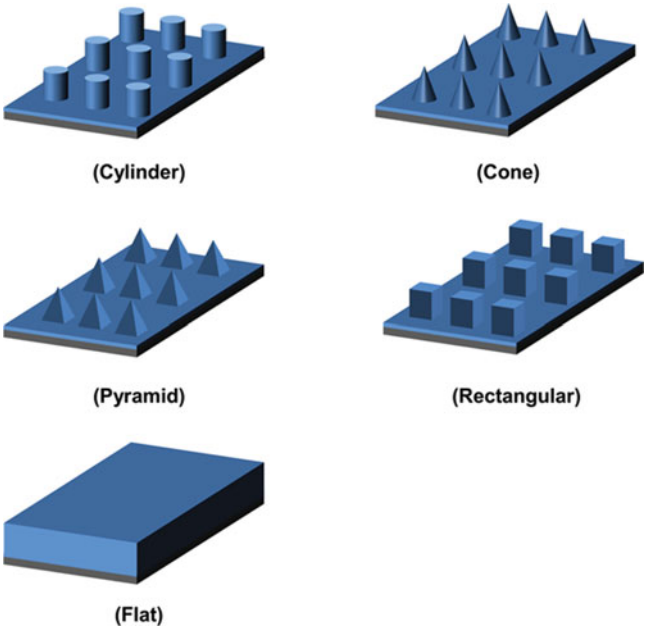


Figure 1. Schematic describing each dielectric structure for interfacial morphologies such as flat, rectangular, cylindrical, pyramidal, and conical.

(cylinder, cone, rectangular, and pyramid) is placed on a dielectric plane that has a thickness of 2 μm. The triboelectric charge density is assumed to be uniformly distributed on the top electrode and on each structure (cylinder, cone, rectangle, and pyramid), including the dielectric plane, and its value is 1×10^{-6} [C/m^2] and -1×10^{-6} [C/m^2], respectively. The dielectric material used here is polytetrafluoroethylene (PTFE), which has a dielectric constant of ~2.

3. Results and discussion

The calculation of transferred charges is explained in this section. Firstly, the distribution of the displacement field on the interface between the dielectric structure and the bottom metal is calculated. This is the induced surface charge density distribution that originates from triboelectric charges on the surface of dielectric materials. The displacement field considers both the electric field and the polarization inside dielectric materials. Under the open-circuit condition, the value of the opposite charges accumulated on the interface is estimated; a sheet of free charge density is formed. In order to consider the morphology effect, it is assumed that an identical quantity of triboelectric charges is uniformly distributed on the surface of each dielectric structure (cone, pyramid, cylinder, rectangle, and flat). Secondly, to calculate the transferred charges on the bottom interface for each structure, the induced surface charge density on the interface is integrated. This corresponds to the quantity of accumulated charges transferred from the top electrode under the short-circuit condition. As a result, the induced charge density on the bottom metal for various structural shapes decreases in the following order: rectangular > cylindrical > pyramidal > conical > flat, as shown in Fig. 2.

Figure 3 shows the in-plane component of the displacement field distribution for various dielectric structures (i.e., flat, rectangular, cylindrical, pyramidal, and conical). The polarity of the induced surface charge density along the x- or y-axis,

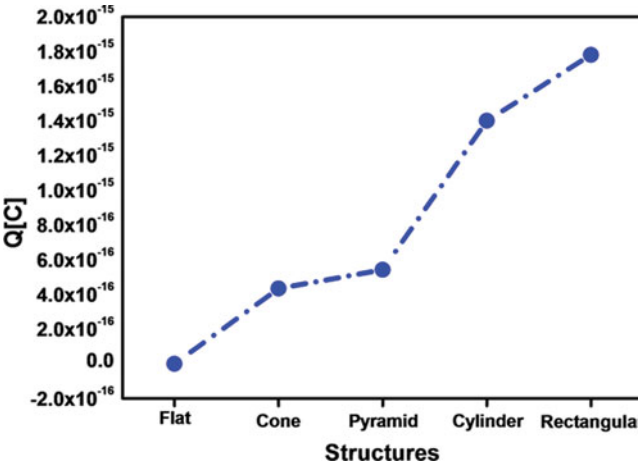


Figure 2. Plot of transferred charges vs. various structures (flat, rectangular, cylindrical, pyramidal, and conical). Note that the rectangular structure shows the highest transferred charges.

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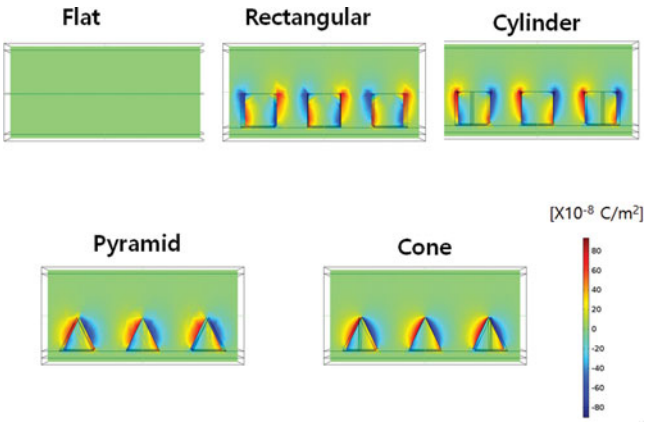


Figure 3. The y (in-plane)-component of the electric displacement field distribution for various interfacial structures (flat, rectangular, cylindrical, pyramidal, and conical).

below and above the surface of the dielectric structures, is seen clearly in Fig. 3, except for the flat structure. The charge density induced on the bottom metal is nearly zero, which means that the in-plane component of the charge density has no impact on the bottom metal.

Figure 4 shows the out-of-plane component of the electric displacement field distribution for various dielectric structures (flat, rectangular, cylindrical, pyramidal, and conical). For the rectangular and cylindrical structures, a large surface charge density is induced on the bottom metal from the triboelectric charges that are uniformly distributed on the surface of the dielectric structures. This estimated induced surface charge density under the open-circuit condition corresponds to the transferred charge density under the short-circuit condition. For the pyramidal and conical structures, the induced surface charge density on the bottom metal is comparatively smaller, and it is the smallest for the flat structure.

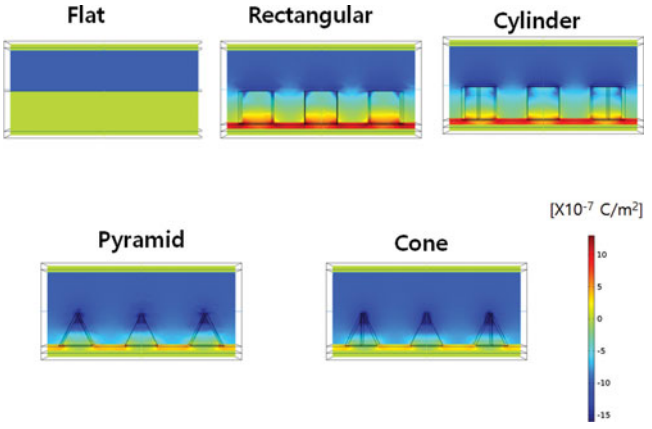


Figure 4. z (out-of-plane)-component of the electric displacement field distribution for various interfacial structures (flat, rectangular, cylindrical, pyramidal, and conical).

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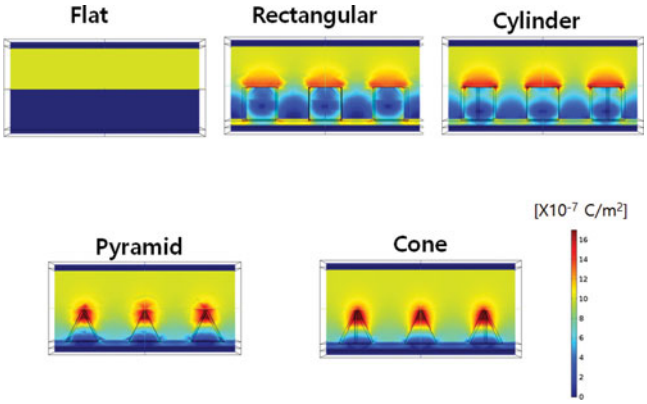


Figure 5. Total electric displacement field distribution for various interfacial structures (flat, rectangular, cylindrical, pyramidal, and conical).

Figure 5 shows the sum of all the components of the displacement field distribution, which is the total induced surface charge density.

$$\|\vec{D}\| = \sqrt{D_x^2 + D_y^2 + D_z^2} \tag{1}$$

where $\vec{D}$ is the displacement field, $\|\|$ is the norm notation, and D_i is the i-th component of $\vec{D}$. The rectangular structure shows the largest induced surface charge density on the bottom metal. Note that calculations show that the surface charge density for the top region of the patterned structure is larger than that for the bottom region. The rectangular structure has the largest surface area among the patterned structures, because of which the largest surface charge density is induced on the bottom metal via electrostatic induction, originating from the triboelectric charges generated on the structure surfaces. One of the factors that affect the electrical output of patterned films over the unstructured films is the transferred charges collected on the bottom electrode, which can be attributed to the current flowing through the external load. The driving force for the transferred charges is the triboelectric charge induced on the structured film. The triboelectric charge density is assumed to be equal for the structured and unstructured films when checking for pure morphological effects (for various interfacial shapes such as rectangular, pyramidal, conical, cylindrical, and flat) numerically using the FEM. The triboelectric effect on the transferred charges of structured films is more than that of flat films with the same thickness. This is because the surface with a larger area, such as a rectangular structure, has a larger and more efficient triboelectric effect, and thus it can generate more transferred charges on the bottom region via electrostatic induction. Compared to the flat structure, in rectangular, cylindrical, conical, and pyramidal structures, the capacitance also changes during the deformation process because of the air voids present. This contributes to the device performance. Owing to the presence of these air voids, structured films with triboelectric charges are detached more easily compared to flat films, which results in easier formation of a potential gradient between the electrodes.

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4. Conclusions

The transferred charges generated by various interfacial structures in a TENG were investigated using an FEM. The simulation results show that understanding the interfacial morphology is an important factor in determining the transferred charges in the TENG. To use the surface structuring effect of the interfacial morphology, a rectangular structure, which has a larger area as compared to other structures, can be applied to obtain large transferred charges. Therefore, the design of interfacial shapes can be considered to provide transferred charges in a controlled manner for self-powered TENG systems.

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Funding

This work was supported by the Office of Computational Energy Science, South Korea.

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